ENGR2001 Engineering Mathematics - Practice Assignment

Section A - Calculus

Section A covers differentiation, integration, and basic differential equations.

Question 1 Differentiate $y = x^3 \cos(2x)$. **[6 marks]**

Question 2 Use logarithmic differentiation to differentiate $y = (x^2 + 1)^{(3x)}$. No marks will be awarded for an alternative method. **[8 marks]**

Question 3 Evaluate integral from 0 to pi/2 of sin(x) dx. **[4 marks]**

Question 4 A current is defined by $I(t) = e^{(-t)} \sin(3t)$. Using $V_L = L\, dI/dt$, derive the inductor voltage. **[10 marks]**

Section B - Complex Numbers and Probability

Section B covers complex numbers, sets, and probability.

Question 5(a) Given $z1 = 3 - 2j$ and $z2 = -1 + 4j$, find $z1 - z2$. **[3 marks]**

Question 5(b) Given $\tanh(x) = 3/4$, express x in terms of a natural logarithm. **[5 marks]**

Question 6 A box contains 30 red components and 20 blue components. Two are selected without replacement. Find the probability that both are red. **[6 marks]**

Question 7 The measured diameters in mm are 10.1, 9.9, 10.0, 10.2, 9.8, 10.0. Calculate the mean and standard deviation. **[8 marks]**

Submission Information

This practice assignment is for revision only. No submission deadline is specified.